

Nazar Nurhilman

22571007

3A Sistem Informasi

Soal!!

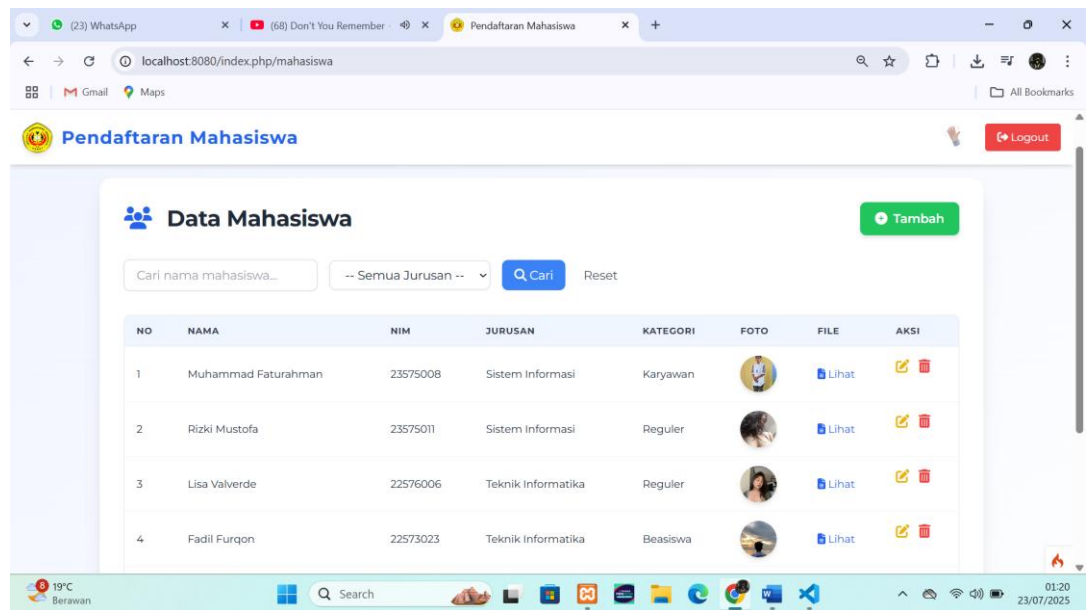
1. Buatlah CRUD sederhana untuk pendaftaran Mahasiswa!
2. Buatlah form handling untuk validasi data mahasiswa
3. Buatlah fitur upload file dan gunakan validasi (PDF, DOCX, JPG, PNG)
4. Buatlah sebuah aplikasi sederhana menggunakan Advanced Query

Jawaban!!

1. CRUD Pendaftaran Mahasiswa:

a. Hasil Tampilan

1) Index



2) Create

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa/create`. The page title is "Pendaftaran Mahasiswa". The main content is a form titled "Tambah Mahasiswa" with the subtitle "Isi data mahasiswa dengan benar". The form fields are as follows:

- Nama:** Arman Hardiansyah
- NIM:** 23575010
- Jurusan:** Teknik Informatika
- Kategori:** Mahasiswa (dropdown menu)
- Foto:** No file chosen (with a "Choose file" button)
- Upload File Baru (PDF, DOCX, 3PG, PNG):** No file chosen (with a "Choose file" button)

At the bottom of the form, there is a "Simpan Data" button and a "Kembali" link. The footer of the page reads "© 2025 Sistem Pendaftaran Mahasiswa | Institut Pendidikan Indonesia Garut".

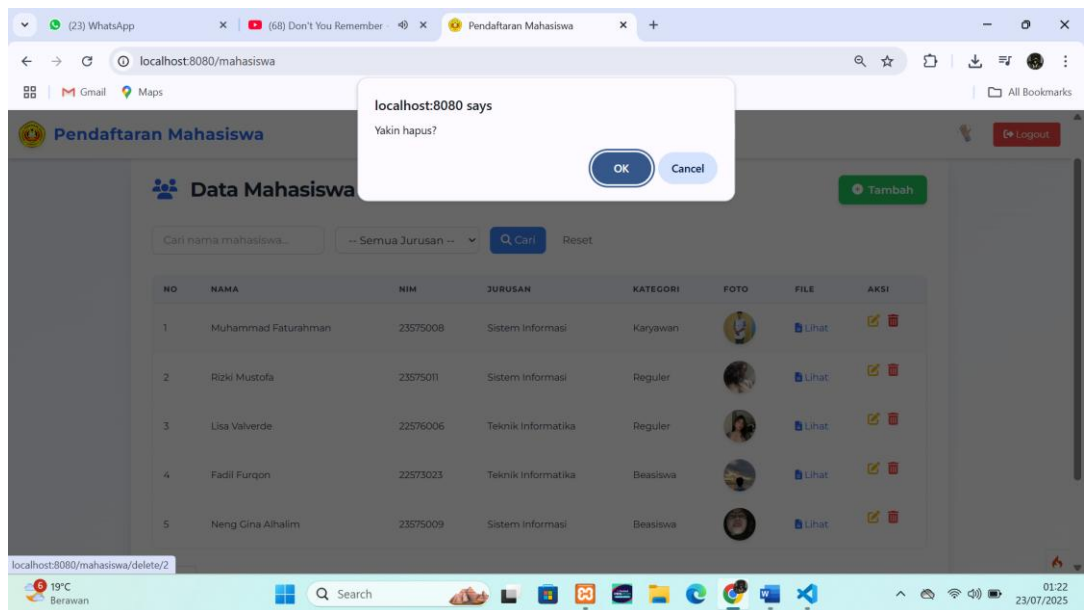
3) Edit

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa/edit/2`. The page title is "Pendaftaran Mahasiswa". The main content is a form titled "Edit Mahasiswa" with the subtitle "Perbarui data mahasiswa dengan benar". The form fields are as follows:

- Nama:** Muhammad Faturrahman
- NIM:** 23575008
- Jurusan:** Sistem Informasi
- Kategori:** Karyawan (dropdown menu)
- Foto Saat Ini:** A small thumbnail image of a person in a white shirt and blue pants.
- Upload File Baru (PDF, DOCX, 3PG, PNG):** No file chosen (with a "Choose file" button)

At the bottom of the form, there is an "Update Data" button and a "Kembali" link. The footer of the page reads "© 2025 Sistem Pendaftaran Mahasiswa | Institut Pendidikan Indonesia Garut".

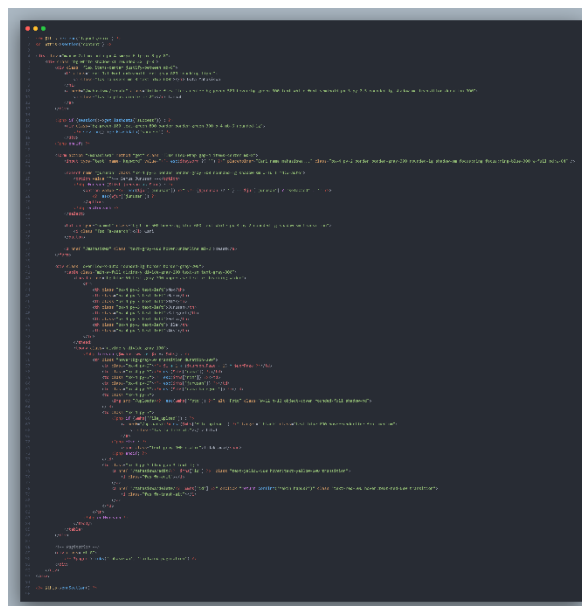
4) Delete



b. Listing Code

1) Index

a) Index.php



b) Controller Index Mahasiswa.php

2) Create

a) Create.php

b) Controller Create Mahasiswa.php

```
1 public function create()
2 {
3     $data['kategori'] = $this->kategoriModel->findAll();
4     return view('mahasiswa/create', $data);
5 }
6
7 public function store()
8 {
9     $validation = \Config\Services::validation();
10    $data = $this->request->getPost();
11
12    $rules = [
13        'nama' => 'required|min_length[3]',
14        'nim' => 'required|numeric|is_unique[mahasiswa.nim]',
15        'jurusan' => 'required',
16        'foto' => 'uploaded[foto]|is_image[foto]',
17        'file_upload' => 'uploaded[file_upload]|ext_in[file_upload,pdf,docx,jpg,png]'
18    ];
19
20    if (!$this->validate($rules)) {
21        return redirect()->back()->withInput()->with('errors', $validation->listErrors());
22    }
23
24    $foto = $this->request->getFile('foto');
25    $namaFoto = $foto->getRandomName();
26    $foto->move('uploads', $namaFoto);
27
28    $file = $this->request->getFile('file_upload');
29    $namaFile = $file->getRandomName();
30    $file->move('uploads', $namaFile);
31
32    $this->mahasiswaModel->save([
33        'nama' => $data['nama'],
34        'nim' => $data['nim'],
35        'jurusan' => $data['jurusan'],
36        'foto' => $namaFoto,
37        'file_upload' => $namaFile,
38        'kategori_id' => $data['kategori_id'] ?? null
39    ]);
40
41    return redirect()->to('/mahasiswa')->with('success', 'data berhasil ditambahkan');
42 }
```

3) Edit

a) Edit.php

```
1 public function edit($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4     $kategori = $this->kategoriModel->findAll();
5
6     if (!$mahasiswa) {
7         return redirect()->to('/mahasiswa')->with('error', 'data tidak ditemukan');
8     }
9
10    return view('mahasiswa/edit', compact('mahasiswa', 'kategori'));
11 }
12
13 public function update($id)
14 {
15     $mahasiswa = $this->mahasiswaModel->find($id);
16     $kategori = $this->kategoriModel->findAll();
17
18     if (!$mahasiswa) {
19         return redirect()->to('/mahasiswa')->with('error', 'data tidak ditemukan');
20     }
21
22     $rules = [
23         'nama' => 'required|min_length[3]',
24         'nim' => 'required|numeric|is_unique[mahasiswa.nim]',
25         'jurusan' => 'required',
26         'foto' => 'uploaded[foto]|is_image[foto]',
27         'file_upload' => 'uploaded[file_upload]|ext_in[file_upload,pdf,docx,jpg,png]'
28     ];
29
30     if (!$this->validate($rules)) {
31         return redirect()->back()->withInput()->with('errors', $validation->listErrors());
32     }
33
34     $foto = $this->request->getFile('foto');
35     $namaFoto = $foto->getRandomName();
36     $foto->move('uploads', $namaFoto);
37
38     $file = $this->request->getFile('file_upload');
39     $namaFile = $file->getRandomName();
40     $file->move('uploads', $namaFile);
41
42     $this->mahasiswaModel->update($id, [
43         'nama' => $data['nama'],
44         'nim' => $data['nim'],
45         'jurusan' => $data['jurusan'],
46         'foto' => $namaFoto,
47         'file_upload' => $namaFile,
48         'kategori_id' => $data['kategori_id'] ?? null
49     ]);
50
51     return redirect()->to('/mahasiswa')->with('success', 'data berhasil diupdate');
52 }
```

b) Controller Edit Mahasiswa.php

```
1 public function edit($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4     $kategori = $this->kategoriModel->findAll();
5
6     return view('mahasiswa/edit', [
7         'mahasiswa' => $mahasiswa,
8         'kategori' => $kategori
9     ]);
10 }
11
12 public function update($id)
13 {
14     $mahasiswa = $this->mahasiswaModel->find($id);
15     $validation = $this->validation->validate($id);
16
17     $rules = [
18         'nama' => 'required|min_length[3]',
19         'nim' => 'required|numeric',
20         'jurusan' => 'required',
21         'file_upload' => 'permit|upload|max_size[5000000]|mime_image'
22     ];
23
24     if (!$this->validation->validate($id)) {
25         return redirect()->back()->withInput()->with('errors', $validation->errors());
26     }
27
28     $data = [
29         'nama' => $this->request->getPost('nama'),
30         'nim' => $this->request->getPost('nim'),
31         'jurusan' => $this->request->getPost('jurusan'),
32         'kategori_id' => $this->request->getPost('kategori_id') // null
33     ];
34
35     $foto = $this->request->getPost('foto');
36     if ($foto && $foto->isValid() && $foto->hasMoved()) {
37         $namaFoto = $foto->getFileName();
38         $foto->move('uploads', $namaFoto);
39         if (file_exists('uploads/' . $mahasiswa['foto'])) {
40             unlink('uploads/' . $mahasiswa['foto']);
41         }
42         $data['foto'] = $namaFoto;
43     }
44
45     $file = $this->request->getPost('file_upload');
46     if ($file && $file->isValid() && $file->hasMoved()) {
47         $namaFile = $file->getFileName();
48         $file->move('uploads', $namaFile);
49         if (file_exists('uploads/' . $mahasiswa['file_upload'])) {
50             unlink('uploads/' . $mahasiswa['file_upload']);
51         }
52         $data['file_upload'] = $namaFile;
53     }
54
55     $this->mahasiswaModel->update($id, $data);
56     return redirect()->to('/mahasiswa')->with('success', 'Data berhasil diupdate');
57 }
```

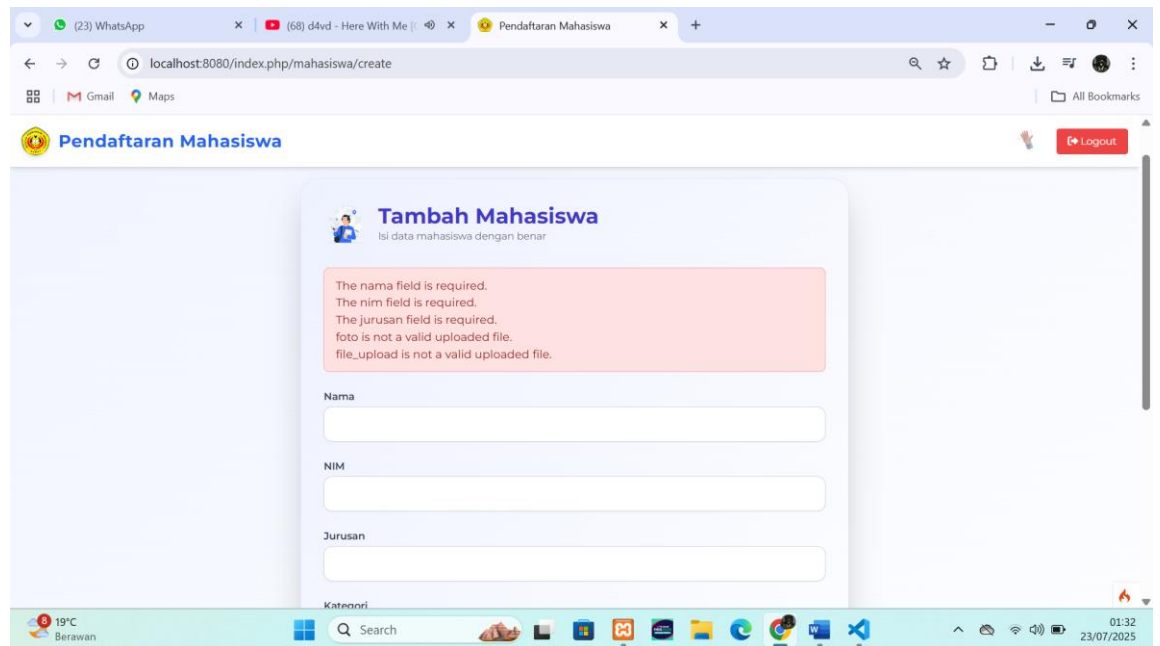
4) Delete

a) Controller Delete Mahasiswa.php

```
1 public function delete($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4
5     if ($mahasiswa) {
6         if (file_exists('uploads/' . $mahasiswa['foto'])) {
7             unlink('uploads/' . $mahasiswa['foto']);
8         }
9         if (file_exists('uploads/' . $mahasiswa['file_upload'])) {
10             unlink('uploads/' . $mahasiswa['file_upload']);
11         }
12     }
13
14     $this->mahasiswaModel->delete($id);
15     return redirect()->to('/mahasiswa')->with('success', 'Data berhasil dihapus.');
```

2. Form Handling + Validasi Data

a. Hasil Tampilan



Note: Seperti tampilan diatas bahwa form create maupun edit sudah menggunakan form handling + validasi, dimana ketika kita tidak mengisi data ada pesan eror dimana nama, nim, jurusan, foto dibutuhkan. Dan nim harus berupa angka jangan huruf.

b. Listing Code

1) Form Handling + Validasi Form Create

```
1 public function create()
2 {
3     $data['kategori'] = $this->kategoriModel->findAll();
4     return view('mahasiswa/create', $data);
5 }
6
7 public function store()
8 {
9     $validation = \Config\Services::validation();
10    $data = $this->request->getPost();
11
12    $rules = [
13        'nama' => 'required|min_length[3]',
14        'nim' => 'required|numeric|is_unique[mahasiswa.nim]',
15        'jurusan' => 'required',
16        'foto' => 'uploaded[foto]|is_image[foto]',
17        'file_upload' => 'uploaded[file_upload]|ext_in[file_upload,pdf,docx,jpg,png]';
18    ];
19
20    if (!$this->validate($rules)) {
21        return redirect()->back()->withInput()->with('errors', $validation->listErrors());
22    }
23
24    $foto = $this->request->getFile('foto');
25    $namaFoto = $foto->getRandomName();
26    $foto->move('uploads', $namaFoto);
27
28    $file = $this->request->getFile('file_upload');
29    $namaFile = $file->getRandomName();
30    $file->move('uploads', $namaFile);
31
32    $this->mahasiswaModel->save([
33        'nama' => $data['nama'],
34        'nim' => $data['nim'],
35        'jurusan' => $data['jurusan'],
36        'foto' => $namaFoto,
37        'file_upload' => $namaFile,
38        'kategori_id' => $data['kategori_id'] ?? null
39    ]);
40
41    return redirect()->to('/mahasiswa')->with('success', 'Data berhasil ditambahkan');
42 }
```

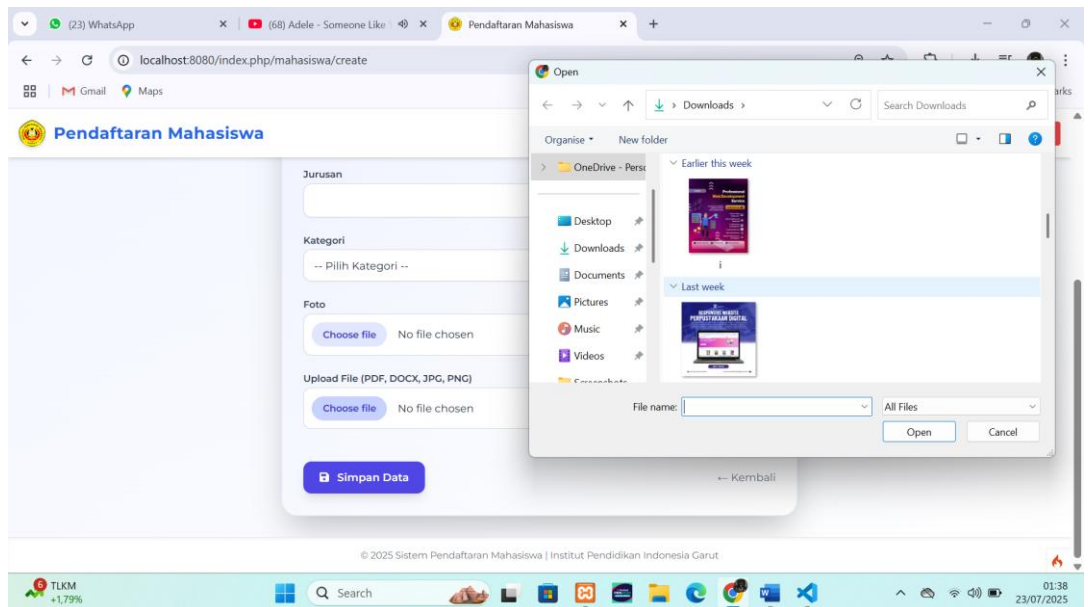
2) Form Handling + Validasi Form Edit

```
1 public function edit($id)
2 {
3     $mahasiswa = $this->mahasiswaModel->find($id);
4     $kategori = $this->kategoriModel->findAll();
5
6     return view('mahasiswa/edit', [
7         'mahasiswa' => $mahasiswa,
8         'kategori' => $kategori
9     ]);
10 }
11
12 public function update($id)
13 {
14     $mahasiswa = $this->mahasiswaModel->find($id);
15     $validation = $this->validation->validation();
16
17     $rules = [
18         'nama' => 'required|min_length(3)',
19         'nim' => 'required|numeric',
20         'jurusan' => 'required',
21         'file_upload' => 'permit|upload|max_size[1024000]|mime_image'
22     ];
23
24     if (!$this->validation->validate($validation)) {
25         return redirect()->back()->withInput()->with('errors', $validation->errors());
26     }
27
28     $data = [
29         'nama' => $this->request->getPost('nama'),
30         'nim' => $this->request->getPost('nim'),
31         'jurusan' => $this->request->getPost('jurusan'),
32         'kategori_id' => $this->request->getPost('kategori_id') // null
33     ];
34
35     $foto = $this->request->getFile('foto');
36     if ($foto && $foto->isValid() && $foto->hasMoved()) {
37         $namaFoto = $foto->getRandomName();
38         $foto->move('uploads', $namaFoto);
39         if ($file_exists('uploads/' . $mahasiswa['foto'])) {
40             unlink('uploads/' . $mahasiswa['foto']);
41         }
42         $data['foto'] = $namaFoto;
43     }
44
45     $file = $this->request->getFile('file_upload');
46     if ($file && $file->isValid() && $file->hasMoved()) {
47         $namaFile = $file->getRandomName();
48         $file->move('uploads', $namaFile);
49         if ($file_exists('uploads/' . $mahasiswa['file_upload'])) {
50             unlink('uploads/' . $mahasiswa['file_upload']);
51         }
52         $data['file_upload'] = $namaFile;
53     }
54
55     $this->mahasiswaModel->update($id, $data);
56     return redirect()->to('/mahasiswa')->with('success', 'Data berhasil diupdate');
57 }
```

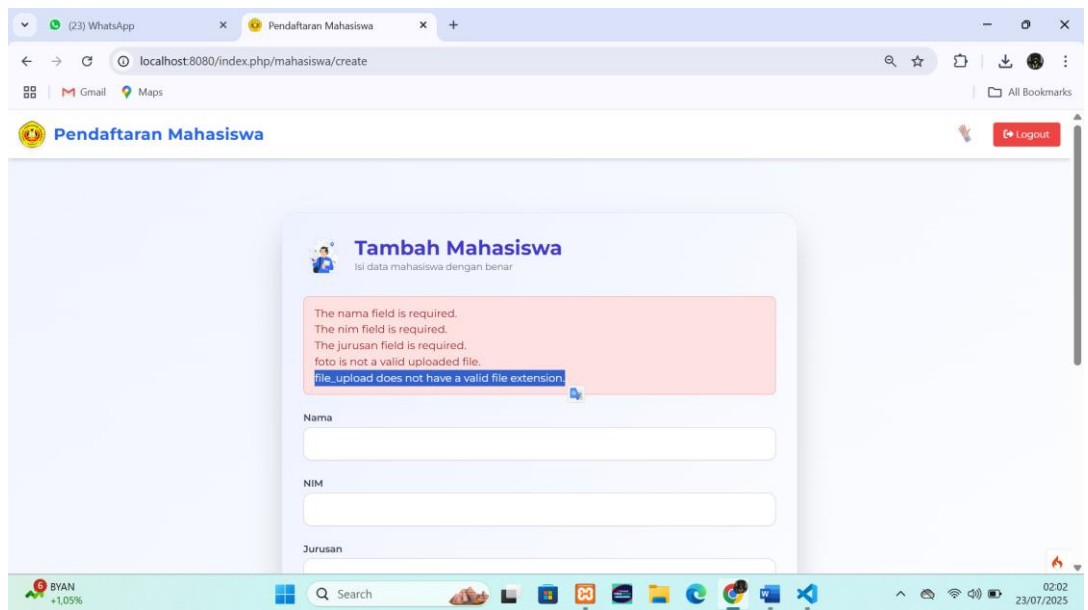
3. Fitur Upload File + Validasi Ektensi File

a. Hasil Tampilan

1) Upload File



2) Validasi File Ektensi



Note: Bisa dilihat seperti digambar bahwa ketika upload file selain ekstensi yang sudah di validasi, maka akan eror dan muncul pesan gagal.

b. Listing Code

```
1  $foto = $this->request->getFile('foto');
2  $namaFoto = $foto->getRandomName();
3  $foto->move('uploads', $namaFoto);
4
5  $file = $this->request->getFile('file_upload');
6  $namaFile = $file->getRandomName();
7  $file->move('uploads', $namaFile);
8
9  $this->mahasiswaModel->save([
10     'nama' => $data['nama'],
11     'nim' => $data['nim'],
12     'jurusan' => $data['jurusan'],
13     'foto' => $namaFoto,
14     'file_upload' => $namaFile,
15     'kategori_id' => $data['kategori_id'] ?? null
16  ]);
```

4. Advance Query

a. Hasil Tampilan

1) Join antar table (kategori)

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa/create`. The page title is "Pendaftaran Mahasiswa". The form contains the following fields:

- Jurusan**: A text input field.
- Kategori**: A dropdown menu with the following options: `-- Pilih Kategori --`, `-- Pilih Kategori --`, **Beasiswa** (selected), `Reguler`, and `Karyawan`.
- Upload File (PDF, DOCX, JPG, PNG)**: A section with a `Choose file` button and the text `No file chosen`.
- Buttons**: `Simpan Data` and `-- Kembali`.

The footer of the page reads: `© 2025 Sistem Pendaftaran Mahasiswa | Institut Pendidikan Indonesia Garut`.

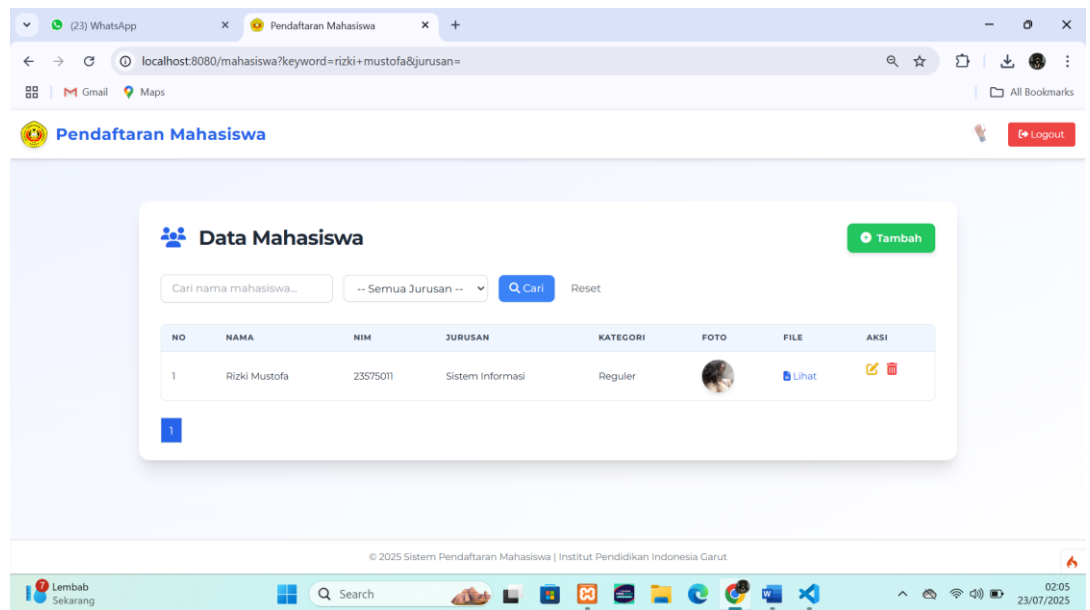
2) Filter Jurusan

The screenshot shows a web browser window with the URL `localhost:8080/mahasiswa?keyword=&jurusan=Sistem+Informasi`. The page title is "Pendaftaran Mahasiswa". The table displays the following data:

NO	NAMA	NIM	JURUSAN	KATEGORI	FOTO	FILE	AKSI
1	Muhammad Faturahman	23575008	Sistem Informasi	Karyawan		Lihat	Edit Hapus
2	Rizki Mustofa	23575011	Sistem Informasi	Reguler		Lihat	Edit Hapus
3	Neng Cina Alhalim	23575009	Sistem Informasi	Beasiswa		Lihat	Edit Hapus
4	Peri Latuyatim	22576004	Sistem Informasi	Reguler		Lihat	Edit Hapus

The footer of the page reads: `Lembab Sekarang`.

3) Filter Nama



b. Listing Code

1) Join Antar Table



2) Filter Jurusan

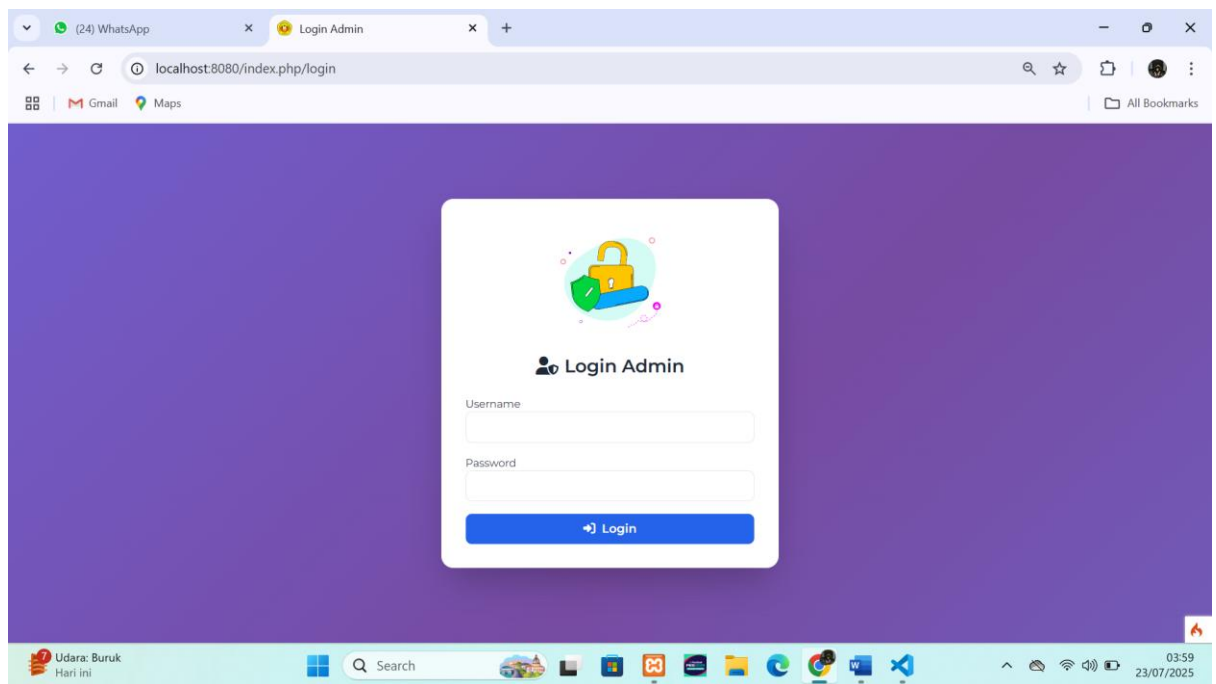


3) Filter Nama

```
1 public function searchByName($keyword)
2 {
3     return $this->like('nama', $keyword)->findAll();
4 }
```

BONUS!!

1. Tampilan Login



2. Listing Code Controller Auth

```
1 <?php
2
3 namespace App\Controllers;
4
5 use App\Models\UserModel;
6 use CodeIgniter\Controller;
7
8 class Auth extends BaseController
9 {
10     public function login()
11     {
12         return view('auth/login');
13     }
14
15     public function loginProcess()
16     {
17         $session = session();
18         $model = new UserModel();
19
20         $username = $this->request->getPost('username');
21         $password = $this->request->getPost('password');
22
23         $user = $model->where('username', $username)->first();
24
25         if ($user && password_verify($password, $user['password'])) {
26             $session->set([
27                 'user_id' => $user['id'],
28                 'username' => $user['username'],
29                 'role' => $user['role'],
30                 'logged_in' => true
31             ]);
32
33             // Langsung redirect ke halaman mahasiswa
34             return redirect()->to('/mahasiswa');
35         } else {
36             return redirect()->back()->with('error', 'Username atau Password salah!');
37         }
38     }
39
40     public function logout()
41     {
42         session()->destroy();
43         return redirect()->to('/login');
44     }
45 }
46
```

3. Listing Code Model

```
1 <?php
2
3 namespace App\Models;
4
5 use CodeIgniter\Model;
6
7 class MahasiswaModel extends Model
8 {
9     protected $table = 'mahasiswa';
10     protected $primaryKey = 'id';
11     protected $allowedFields = [
12         'nama', 'nim', 'jurusan', 'foto', 'file_upload', 'kategori_id', 'created_at'
13     ];
14
15     // Optional: jika ingin pakai timestamp otomatis
16     // MahasiswaModel.php
17     protected $useTimestamps = false;
18
19
20     // Join ke tabel kategori
21     public function getWithKategori()
22     {
23         return $this->select('mahasiswa.*', kategori.nama_kategori')
24             ->join('kategori', 'kategori.id = mahasiswa.kategori_id', 'left')
25             ->findAll();
26     }
27
28     // WHERE Query: Filter berdasarkan jurusan
29     public function filterByJurusan($jurusan)
30     {
31         return $this->where('jurusan', $jurusan)->findAll();
32     }
33
34     // LIKE Query: Cari berdasarkan nama
35     public function searchByName($keyword)
36     {
37         return $this->like('nama', $keyword)->findAll();
38     }
39
40     // LIMIT Query: Ambil data terakhir sejumlah limit
41     public function getLatest($limit = 5)
42     {
43         return $this->orderBy('created_at', 'DESC')->limit($limit)->find();
44     }
45
46     // GROUP BY Query: Hitung jumlah mahasiswa per jurusan
47     public function countByJurusan()
48     {
49         return $this->select('jurusan, COUNT(*) as total')
50             ->groupBy('jurusan')
51             ->findAll();
52     }
53 }
54
```

4. Listing Code Route

```

1  <?php
2
3  use CodeIgniter\Router\RouteCollection;
4
5  /**
6   * @var RouteCollection $routes
7   */
8  $routes->get('/', 'Auth::login');
9  $routes->get('/login', 'Auth::login');
10 $routes->post('/login', 'Auth::loginProcess');
11 $routes->get('/logout', 'Auth::logout');
12
13 $routes->group('/', ['filter' => 'auth'], function($routes) {
14     $routes->get('/dashboard', 'Mahasiswa::dashboard');
15     // CRUD Mahasiswa
16     $routes->get('/mahasiswa', 'Mahasiswa::index');
17     $routes->get('/mahasiswa/create', 'Mahasiswa::create');
18     $routes->post('/mahasiswa/store', 'Mahasiswa::store');
19     $routes->get('/mahasiswa/edit/{:num}', 'Mahasiswa::edit/{:num}');
20     $routes->post('/mahasiswa/update/{:num}', 'Mahasiswa::update/{:num}');
21     $routes->get('/mahasiswa/delete/{:num}', 'Mahasiswa::delete/{:num}');
22     $routes->get('/mahasiswa/search', 'Mahasiswa::search');
23     $routes->get('/mahasiswa/jurusan/{:any}', 'Mahasiswa::jurusan/{:any}');
24 });
25
26
27
28

```

5. Listing Code Login.php

[illegible]

Link Repository Github: https://github.com/RizkiMs100404/pendaftaran_mahasiswa